Experiment 8

AIM: Develop Inventory Management Module using ADO.NET & CRUD Operations

Code:

Database\

InventoryDb.sql:

```
-- Database Schema and Table Creation Script
CREATE DATABASE InventoryDb;
GO

USE InventoryDb;
GO

CREATE TABLE InventoryItems (
    Id INT IDENTITY(1,1) PRIMARY KEY,
    ItemName NVARCHAR(100) NOT NULL,
    Category NVARCHAR(50) NOT NULL,
    Quantity INT NOT NULL,
    UnitPrice DECIMAL(10,2) NOT NULL,
    CreatedDate DATETIME DEFAULT GETDATE()
);

-- Seed Initial Inventory Records
INSERT INTO InventoryItems (ItemName, Category, Quantity, UnitPrice, CreatedDate)
VALUES
('Dell Latitude 3420 Laptop', 'Computers', 25, 52000.00, GETDATE()),
('Logitech MX Master 3S Mouse', 'Accessories', 60, 7999.00, GETDATE()),
('Mechanical Gaming Keyboard', 'Accessories', 45, 3499.00, GETDATE()),
('HP LaserJet Pro Printer', 'Peripherals', 12, 18500.00, GETDATE()),
('Samsung 27-inch 4K Monitor', 'Displays', 30, 24999.00, GETDATE());
```

Models\

InventoryItem.cs:

```
using System.ComponentModel.DataAnnotations;

namespace InventoryManagementADONET.Models;

public class InventoryItem
{
    public int Id { get; set; }

    [Required(ErrorMessage = "Item Name is required.")]
    [StringLength(100, ErrorMessage = "Item Name cannot exceed 100 characters.")]
    [Display(Name = "Item Name")]
    public string ItemName { get; set; } = string.Empty;

    [Required(ErrorMessage = "Category is required.")]
    [StringLength(50)]
    public string Category { get; set; } = string.Empty;

    [Required(ErrorMessage = "Quantity is required.")]
    [Range(0, 100000, ErrorMessage = "Quantity must be a positive integer.")]
    public int Quantity { get; set; }

    [Required(ErrorMessage = "Unit Price is required.")]
    [Range(0.01, 1000000.00, ErrorMessage = "Unit Price must be greater than zero.")]
    [Display(Name = "Unit Price (INR)")]
    public decimal UnitPrice { get; set; }

    [Display(Name = "Stock Valuation")]
    public decimal TotalValue => Quantity * UnitPrice;

    [Display(Name = "Date Added")]
    public DateTime CreatedDate { get; set; } = DateTime.Now;
}
```

Data\

InventoryDAL.cs:

```
using System.Data;
using Microsoft.Data.SqlClient;
using InventoryManagementADONET.Models;

namespace InventoryManagementADONET.Data;

public class InventoryDAL
{
    private readonly string _connectionString;

    public InventoryDAL(string connectionString)
    {
        _connectionString = connectionString;
    }

    // SELECT ALL (READ)
    public List<InventoryItem> GetAllItems()
    {
        var items = new List<InventoryItem>();
        using var connection = new SqlConnection(_connectionString);
        using var command = new SqlCommand("SELECT Id, ItemName, Category, Quantity, UnitPrice,
     CreatedDate FROM InventoryItems ORDER BY Id DESC", connection);

        connection.Open();
        using var reader = command.ExecuteReader();
        while (reader.Read())
        {
            items.Add(new InventoryItem
            {
                Id = reader.GetInt32(0),
                ItemName = reader.GetString(1),
                Category = reader.GetString(2),
```

```
                Quantity = reader.GetInt32(3),
                UnitPrice = reader.GetDecimal(4),
                CreatedDate = reader.GetDateTime(5)
            });
        }
        return items;
}

// SELECT BY ID (READ SINGLE)
public InventoryItem? GetItemById(int id)
{
    using var connection = new SqlConnection(_connectionString);
    using var command = new SqlCommand("SELECT Id, ItemName, Category, Quantity, UnitPrice,
 CreatedDate FROM InventoryItems WHERE Id = @Id", connection);
    command.Parameters.Add(new SqlParameter("@Id", SqlDbType.Int) { Value = id });

    connection.Open();
    using var reader = command.ExecuteReader();
    if (reader.Read())
    {
        return new InventoryItem
        {
            Id = reader.GetInt32(0),
            ItemName = reader.GetString(1),
            Category = reader.GetString(2),
            Quantity = reader.GetInt32(3),
            UnitPrice = reader.GetDecimal(4),
            CreatedDate = reader.GetDateTime(5)
        };
    }
    return null;
}
```

```
// INSERT (CREATE)
public bool InsertItem(InventoryItem item)
{
    using var connection = new SqlConnection(_connectionString);
    using var command = new SqlCommand(
        "INSERT INTO InventoryItems (ItemName, Category, Quantity, UnitPrice, CreatedDate)
VALUES (@ItemName, @Category, @Quantity, @UnitPrice, @CreatedDate)",
        connection);

    command.Parameters.Add(new SqlParameter("@ItemName", SqlDbType.NVarChar, 100) { Value =
 item.ItemName });
    command.Parameters.Add(new SqlParameter("@Category", SqlDbType.NVarChar, 50) { Value =
item.Category });
    command.Parameters.Add(new SqlParameter("@Quantity", SqlDbType.Int) { Value = item.Quan
tity });
    command.Parameters.Add(new SqlParameter("@UnitPrice", SqlDbType.Decimal) { Value = item
.UnitPrice });
    command.Parameters.Add(new SqlParameter("@CreatedDate", SqlDbType.DateTime) { Value = i
tem.CreatedDate });

    connection.Open();
    return command.ExecuteNonQuery() > 0;
}

// UPDATE
public bool UpdateItem(InventoryItem item)
{
    using var connection = new SqlConnection(_connectionString);
    using var command = new SqlCommand(
        "UPDATE InventoryItems SET ItemName = @ItemName, Category = @Category, Quantity = @
Quantity, UnitPrice = @UnitPrice WHERE Id = @Id",
        connection);

    command.Parameters.Add(new SqlParameter("@Id", SqlDbType.Int) { Value = item.Id });
```

```
        command.Parameters.Add(new SqlParameter("@ItemName", SqlDbType.NVarChar, 100) { Value =
     item.ItemName });
        command.Parameters.Add(new SqlParameter("@Category", SqlDbType.NVarChar, 50) { Value =
    item.Category });
        command.Parameters.Add(new SqlParameter("@Quantity", SqlDbType.Int) { Value = item.Quan
    tity });
        command.Parameters.Add(new SqlParameter("@UnitPrice", SqlDbType.Decimal) { Value = item
    .UnitPrice });

        connection.Open();
        return command.ExecuteNonQuery() > 0;
    }

    // DELETE
    public bool DeleteItem(int id)
    {
        using var connection = new SqlConnection(_connectionString);
        using var command = new SqlCommand("DELETE FROM InventoryItems WHERE Id = @Id", connect
    ion);
        command.Parameters.Add(new SqlParameter("@Id", SqlDbType.Int) { Value = id });

        connection.Open();
        return command.ExecuteNonQuery() > 0;
    }
}
```

Controllers\

InventoryController.cs:

```
using Microsoft.AspNetCore.Mvc;
using InventoryManagementADONET.Data;
using InventoryManagementADONET.Models;

namespace InventoryManagementADONET.Controllers;
```

```
public class InventoryController : Controller
{
    private readonly InventoryDAL _dal;

    public InventoryController(IConfiguration configuration)
    {
        var connStr = configuration.GetConnectionString("DefaultConnection")
                      ?? "Server=.;Database=InventoryDb;Trusted_Connection=True;TrustServerCert
    ificate=True;";
        _dal = new InventoryDAL(connStr);
    }

    public IActionResult Index()
    {
        var items = _dal.GetAllItems();
        return View(items);
    }

    [HttpGet]
    public IActionResult Create()
    {
        return View(new InventoryItem());
    }

    [HttpPost]
    [ValidateAntiForgeryToken]
    public IActionResult Create(InventoryItem item)
    {
        if (!ModelState.IsValid) return View(item);

        item.CreatedDate = DateTime.Now;
        if (_dal.InsertItem(item))
        {
            TempData["SuccessMessage"] = $"Item '{item.ItemName}' added successfully.";
```

```
            return RedirectToAction(nameof(Index));
        }

        ModelState.AddModelError("", "Database error occurred during insertion.");
        return View(item);
    }

    [HttpGet]
    public IActionResult Edit(int id)
    {
        var item = _dal.GetItemById(id);
        return item == null ? NotFound() : View(item);
    }

    [HttpPost]
    [ValidateAntiForgeryToken]
    public IActionResult Edit(int id, InventoryItem item)
    {
        if (id != item.Id) return BadRequest();
        if (!ModelState.IsValid) return View(item);

        if (_dal.UpdateItem(item))
        {
            TempData["SuccessMessage"] = $"Item '{item.ItemName}' updated successfully.";
            return RedirectToAction(nameof(Index));
        }

        ModelState.AddModelError("", "Database error occurred during update.");
        return View(item);
    }

    [HttpGet]
    public IActionResult Delete(int id)
    {
        var item = _dal.GetItemById(id);
```

```
        return item == null ? NotFound() : View(item);
    }

    [HttpPost, ActionName("Delete")]
    [ValidateAntiForgeryToken]
    public IActionResult DeleteConfirmed(int id)
    {
        var item = _dal.GetItemById(id);
        string name = item?.ItemName ?? "Item";
        if (_dal.DeleteItem(id))
        {
            TempData["SuccessMessage"] = $"Item '{name}' deleted from inventory.";
        }
        return RedirectToAction(nameof(Index));
    }
}
```

Views\Inventory\

Index.cshtml:

```
@model IEnumerable<InventoryManagementADONET.Models.InventoryItem>
@{
    ViewData["Title"] = "Hardware & Accessories Inventory";
}

<div class="d-flex justify-content-between align-items-center mb-4">
    <div>
        <span class="badge bg-success text-uppercase px-2 py-1">ADO.NET CRUD</span>
        <h2 class="mt-2 fw-bold text-dark">Hardware &amp;amp; Accessories Inventory</h2>
        <p class="text-muted mb-0">Direct SQL operations using safe parameterized queries and S
    qlDataReader</p>
    </div>
    <a asp-action="Create" class="btn btn-primary px-3 py-2 shadow-sm">
        Add Inventory Item
```

```
    </a>
</div>

<div class="card shadow-sm border-0 rounded-3">
    <div class="card-body p-0">
        <div class="table-responsive">
            <table class="table table-hover align-middle mb-0">
                <thead class="table-light">
                    <tr>
                        <th class="ps-4">Item ID</th>
                        <th>Item Name</th>
                        <th>Category</th>
                        <th>Stock Qty</th>
                        <th>Unit Price</th>
                        <th>Total Valuation</th>
                        <th>Created Date</th>
                        <th class="text-center pe-4">Actions</th>
                    </tr>
                </thead>
                <tbody>
                    @foreach (var item in Model)
                    {
                        <tr>
                            <td class="ps-4 fw-bold text-muted">#@item.Id</td>
                            <td class="fw-bold text-dark">@item.ItemName</td>
                            <td><span class="badge bg-secondary">@item.Category</span></td>
                            <td><span class="badge bg-success">@item.Quantity units</span></td>
                            <td>&#8377; @item.UnitPrice.ToString("N2")</td>
                            <td class="fw-bold text-primary">&#8377; @item.TotalValue.ToString(
    "N2")</td>
                            <td>@item.CreatedDate.ToString("dd MMM yyyy")</td>
                            <td class="text-center pe-4">
                                <a asp-action="Edit" asp-route-id="@item.Id" class="btn btn-sm
```

```
    btn-outline-primary">Edit</a>
                                <a asp-action="Delete" asp-route-id="@item.Id" class="btn btn-s
    m btn-outline-danger">Delete</a>
                            </td>
                        </tr>
                    }
                </tbody>
            </table>
        </div>
    </div>
</div>
```

Views\Inventory\

Create.cshtml:

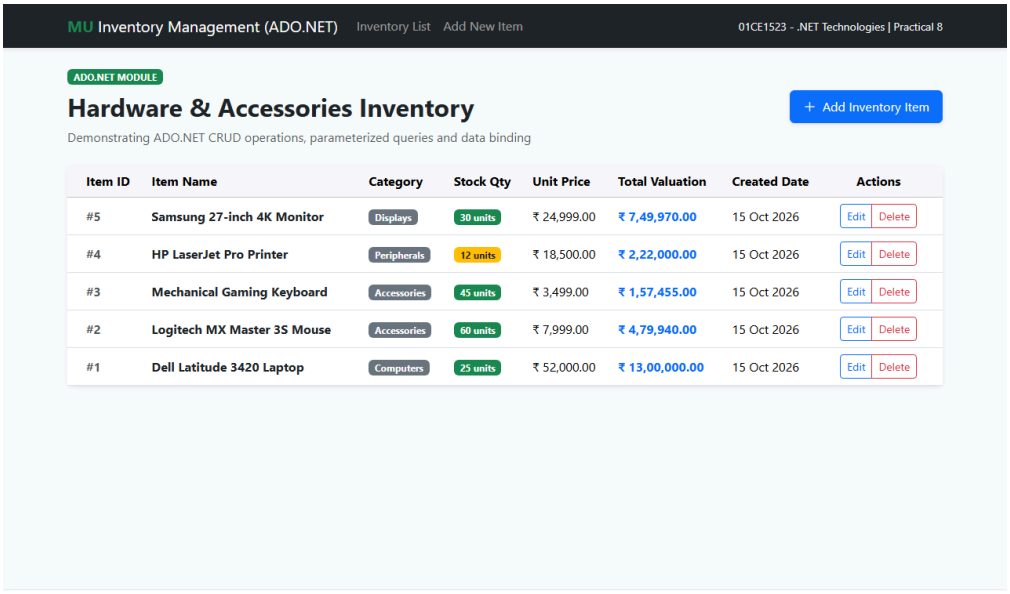
```
@model InventoryManagementADONET.Models.InventoryItem
@{
    ViewData["Title"] = "Add Inventory Item";
}

<div class="row justify-content-center">
    <div class="col-lg-7">
        <div class="card shadow-sm border-0 rounded-3">
            <div class="card-header bg-primary text-white py-3">
                <h5 class="mb-0 fw-bold">Add New Inventory Item</h5>
                <small>ADO.NET Parameterized INSERT Query</small>
            </div>
            <div class="card-body p-4">
                <form asp-action="Create" method="post">
                    @Html.AntiForgeryToken()
                    <div class="mb-3">
                        <label asp-for="ItemName" class="form-label fw-bold"></label>
                        <input asp-for="ItemName" class="form-control" placeholder="e.g. Dell M
```

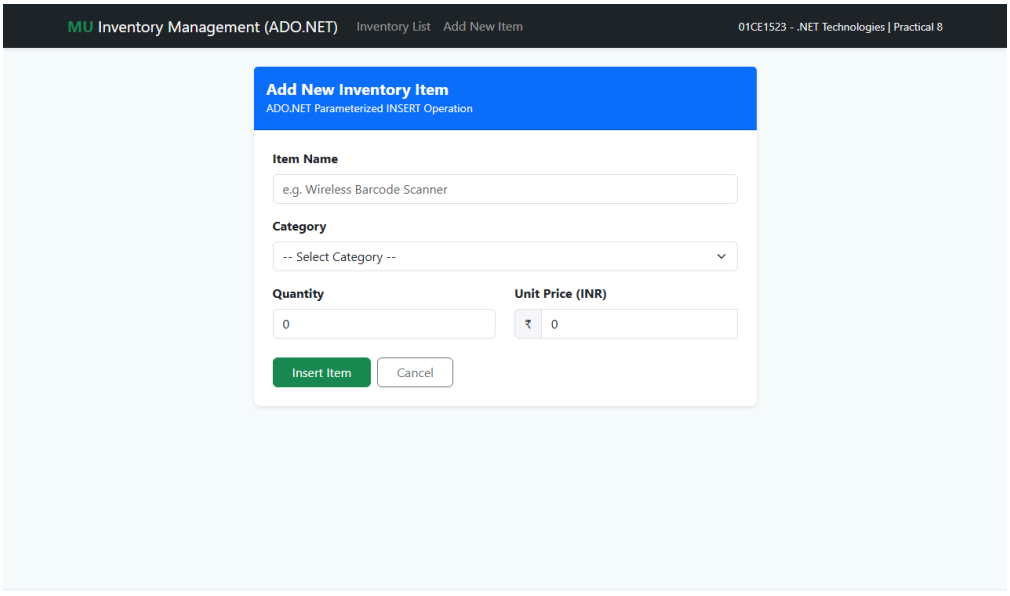
```
onitor 27-inch" />
                    <span asp-validation-for="ItemName" class="text-danger small"></span>
                </div>
                <div class="mb-3">
                    <label asp-for="Category" class="form-label fw-bold"></label>
                    <select asp-for="Category" class="form-select">
                        <option value="Computers">Computers</option>
                        <option value="Accessories">Accessories</option>
                        <option value="Peripherals">Peripherals</option>
                        <option value="Displays">Displays</option>
                    </select>
                </div>
                <div class="row mb-3">
                    <div class="col-md-6">
                        <label asp-for="Quantity" class="form-label fw-bold"></label>
                        <input asp-for="Quantity" type="number" class="form-control" />
                    </div>
                    <div class="col-md-6">
                        <label asp-for="UnitPrice" class="form-label fw-bold"></label>
                        <input asp-for="UnitPrice" type="number" step="0.01" class="form-co
ntrol" />
                    </div>
                </div>
                <div class="d-flex gap-2">
                    <button type="submit" class="btn btn-success px-4">Insert Item</button>
                    <a asp-action="Index" class="btn btn-outline-secondary px-4">Cancel</a>
                </div>
            </form>
        </div>
    </div>
</div>
```

Output: Screenshot:

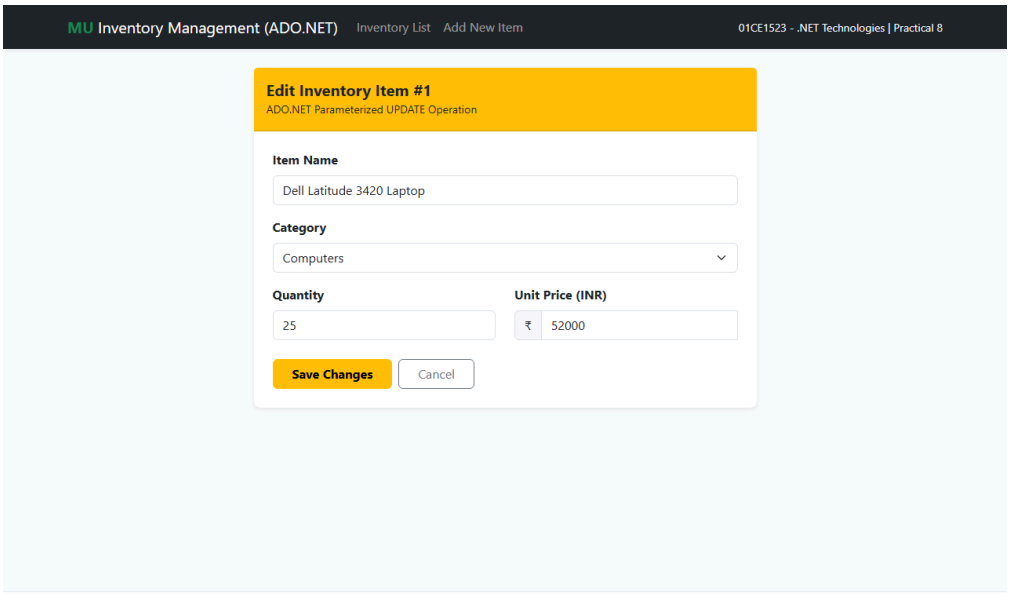
https://localhost:5001/Inventory



https://localhost:5001/Inventory/Create

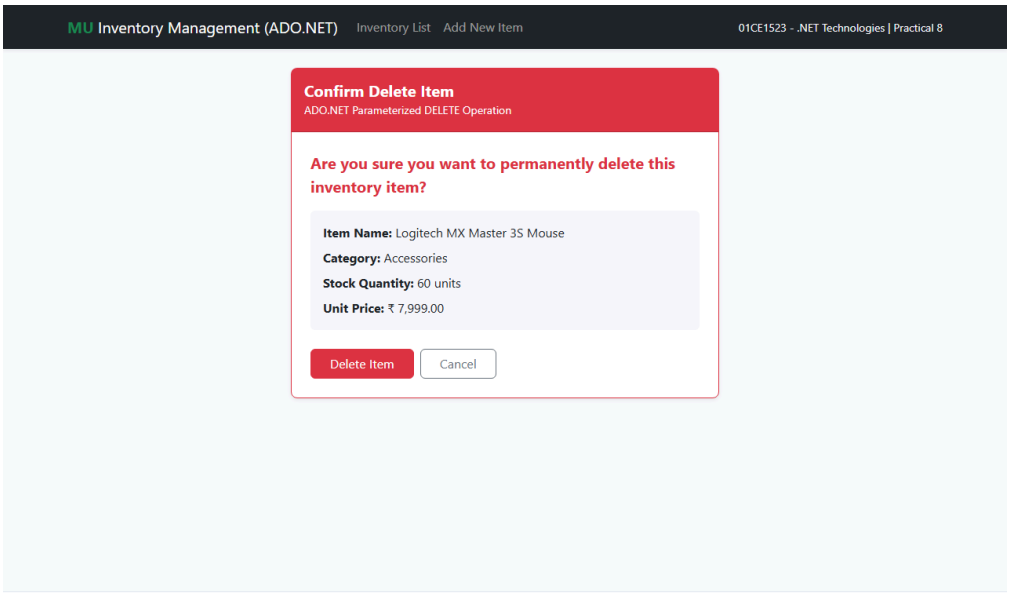


https://localhost:5001/Inventory/Edit/1



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https://localhost:5001/Inventory/Delete/2



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GitHub URL: https://github.com/Anubhavg1123/Practical-8